5 The table shows information about the weekly earnings of 20 people who work in a shop.

Midpoint	Weekly earnings (£x)	Frequency	$f \times$ midpoint
200	$150 < x \leqslant 250$	1	200
300	$250 < x \leqslant 350$	11	3300
400	$350 < x \leqslant 450$	5	2000
500	$450 < x \leqslant 550$	0	0
600	$550 < x \leqslant 650$	3	1800

(a) Work out an estimate for the mean of the weekly earnings.

1 mark
Use each class midpoint and multiply by frequency

Total estimated earnings = 200 + 3300 + 2000 + 0 + 1800 = 7300

Estimated mean = 7300 ÷ 20 = 730 ÷ 2.

$$2 \overline{)\, 7^{1}3^{1}0} \quad 3\ 6\ 5$$

1 mark
Divide by the total frequency, 20

1 mark
Correct estimated mean

£ 365

(3)

Nadiya says,

"The mean may **not** be the best average to use to represent this information."

(b) Do you agree with Nadiya?
You must justify your answer.

Yes. The very high earnings pull the mean upwards.

The median would be less affected by these extreme values.

(1)

1 mark
Explain the effect of extreme earnings

(Total for Question 5 is 4 marks)